Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I

1 Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I

Study focus: Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000250 · EV0000251

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and in today's class we are going to study about corrosion control.

Key details

- We will look at another method of corrosion control and that is using corrosion inhibitors.
- There are two types of corrosion inhibitors, anodic corrosion inhibitors and cathodic corrosion inhibitors.
- They are chemical substances which are added in small quantities to the corrosive environment in order to decrease the rate of corrosion.
- Environment and based on that, and also the nature of the metal, which is which needs to be protected, we are going to choose a corrosion inhibitor and use it, add it to the corrosive environment, and this should be added only in small quantities.
- Now because we know there are two different reactions which happen at dianode and cathode, so there would be different types of corrosion inhibitors.
- Basically, they are going to inhibit either the anodic reaction or they'll inhibit the cathodic reaction and therefore they will help us to control corrosion.
- But the only.
- Problem with this corrosion inhibitor is that it can be used only in a confined environment.
- That can be understood because the concept is we are adding some substance into the corrosive environment.

- It has to be an enclosed environment so we say it is used only in confined environment.
- Action of the corrosion inhibitor obviously will depend on the article which has to be protected as well as the corrosive environment.
- We need to study the specific case.
- what is the nature of the metal, what is the corrosive environment, and then decide what corrosion inhibitor would be suitable to protect that article from corrosion.
- We know metal is getting oxidized forming mn plus n electron n electrons.
- If we achieve that, then we are going to retard the corrosion process.
- This is done with the help of addition of large anions.
- It requires large anions because large anions like chromatic usually will form precipitate with these metal ions.
- What is the reason for that?
- We know that the metal surface is in the is acting as the anode.
- They are going to form a precipitate as I said just now, and they are going to cover the surface of the anode.
- Now, if this covering is not uniform, if there are gaps, then your anode is getting exposed, and therefore this will.

Example or application from the lecture

- Let us understand what exactly are corrosion inhibitors.
- Let us understand what is an anodic inhibitor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 05:00–09:59

small amount of the anode area is getting exposed because the precipitate is not completely covering the.

Key details

- Surface and therefore this will lead to intense corrosion.
- So, therefore, whenever anodic inhibitors are used, we have to make sure that we add sufficient amount of the inhibitor.
- Inhibitor we are using here so sodium chromate sodium tungstate.
- Now when we come to the cathode, we know from the electrochemical theory of corrosion that there can be different types of reactions which will take place because the metal itself is not undergoing reduction.
- It cannot undergo reduction.
- And what were the two types of reactions?
- we had classified into they were cathodic reactions were either it could be liberation of hydrogen or it could be absorption of high oxygen.
- So that means if we want to control corrosion, then we need to look at ways by which either we can stop the liberation of hydrogen or we can stop the absorption of oxygen.

- Now because the cathodic reaction has become slow, it is going to in turn slow down the anodic reaction.
- And once anodic reaction is not taking place, we know that the corrosion is under control or you have inhibit corrosion to a good extent.
- How do we inhibit the liberation of hydrogen?
- It can be done in two different ways.
- This reaction is happening at the cathode and one of the reasons for the reaction would be that H plus ions are being reduced at the cathode.
- Organic compounds containing nitrogen and sulfur.
- How are they doing it?
- All these are widely used as corrosion inhibitors and what is their role?
- They are going to prevent the diffusion of H plus ions to the cathode.
- They get absorbed on the cathodic region they get absorbed on the cathodic surface and form a protective film.
- Another way of doing it is also liberation of hydrogen can be prevented by increasing the hydrogen overvoltage on the metal surface.
- Now when we studied factors affecting corrosion rate, one important factor which we studied was hydrogen over voltage and what did we understand there that if the hydrogen over voltage is high then the liberation of hydrogen is going to be less and therefore we can control we can reduce the rate of reaction or the rate of.

Example or application from the lecture

- First, let us take up hydrogen liberation.
- Let us study the first one where we are going to prevent the diffusion of H plus ions to the cathode.
- First let us look at some examples.

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Core explanation · 10:00–14:59

Corrosion is going to be less.

Key details

- What we have to do is we have to increase the hydrogen over voltage over the cathodic surface.
- It is done with the help of oxides of arsenic and antimony.
- What happens is they are going to deposit arsenic and antimony onto the cathodic region.
- Arsenic and antimony have very high.
- Hydrogen over voltage.
- Now since you have raised the hydrogen over voltage of that particular surface, the liberation of hydrogen becomes difficult and therefore corrosion is going to the rate of corrosion is going to fall down.
- How do we inhibit oxygen absorption?

- And that should not be very difficult.
- All you have to do is oxygen has to be removed.
- It is going to react with the oxygen and it will form nitrogen and H2O.
- You would have removed the oxygen from the atmosphere.
- What's a scavenger?
- It is going to eat up the oxygen.
- So oxygen scavenger is being used here, it's going to take away all the oxygen from the atmosphere, and therefore, when there is no oxygen, obviously oxygen absorption will not.
- happen meaning oxygen is not available to undergo the reduction reaction and hence you would have brought down the extent of cathodic reaction.
- and therefore we can understand that there is a loss of electrons and therefore now we know that the reducing agents when we add it is going to take up the oxygen and that will help us to remove oxygen from the atmosphere and therefore it will reduce the cathodic reaction.
- Another way of reducing or removing the oxygen is or reducing oxygen absorption is by decreasing the diffusion rate of oxygen to the cathode.
- We are using large cations like zinc sulfate, zinc and magnesium.
- So the salts used are zinc sulphate and magnesium sulphate nickel sulphate etc why these salt because their cations are large zinc and mg2 plus zinc 2 plus mg2 plus now these are going to migrate towards the cathodic surface and react with the hydroxyl ions.
- Again this is a precipitate.
- Magnesium also reacts with hydroxide and it forms zinc.
- Magnesium hydroxide which is a precipitate.
- Now these hydroxides again they are getting deposited over on the cathodic sites and when that happens a protective film is formed over the cathodic sites and therefore oxygen cannot come in contact with the cathode and take up electrons and this is how you are preventing the absorption of oxygen.
- The anodic inhibitors and the cathodic inhibitors, how they act.

Example or application from the lecture

- Arsenic and antimony oxide, say for example, these will be the corrosion inhibitors.
- Let us talk of oxygen absorption.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 15:00–19:59

In the case of cathodic inhibitors, depending on which reaction we are going to stop, if it is liberation of hydrogen, then we use urea, thiourea.

Key details

- And another was to increase the hydrogen over voltage that was also done with the help of arsenic oxide and antimony oxide.
- And there was another set of cathodic inhibitors where you could use oxygen scavengers like hydrazine or a reducing agent like sodium sulfide and also if you want to prevent oxygen

from coming in contact with the cathode surface we could use zinc sulfate, magnesium sulphate etc.
- Now this is important, they will contaminate the environment because we know that corrosion inhibitors are substances we are adding directly into the environment.
- Now these are toxic and therefore they cannot be used in systems which come in contact with humans.
- And the last fall, which we have already understood, is that they can be used only in closed system.
- We cannot use them in open system, but only in closed system in which the corrosion corrosive environment is either.
- Contained or it is recirculated.
- Nevertheless, we have to study the nature of the metal, the and corrosive environment, and then decide what type of corrosion inhibitor we are going to use for that particular system.
- So that is about corrosion inhibitors and in the next class we are going to talk about another method of corrosion control and that is cathodic protection.

Example or application from the lecture

- And many of these inhibitors are toxic, like for example arsenic oxide.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000262 · EV0000263 · EV0000264 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9964; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...

C Revision and retrieval practice